

GITCG Universal Protocol V1.0

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1 Introduction

GITCG Universal Protocol, also known as *GIUP*, is a descriptive definition of game Genius Invokation TCG¹. GIUP is committed to creating a universal, transferable definition of situations and actions for GITCG. GIUP isn't a rulebook for GITCG - it's not trying to be the boss here. Instead, GIUP's goal is to describe the current state of play in GITCG and the potential action space, without getting tangled up in how different elements interact. So, we claim it *descriptive*.

2 Terminology

This section will introduce the key terms in GITCG and provide precise definitions.

Term 1: Player

GIUP uses a number starting from zero to represent players. In the basic version, there are only two options: 0 or 1. In the extended version, there will be other options, such as a random player (2).

Term 2: Current Player

Current Player represents the one who can take an action at now.

Term 3: Round

It is exactly same as Round in the game.

Term 4: Turn

When current player changes from 0 to 1 or 1 to 0, Turn +1. At the beginning of each Round, Turn is set to 1.

This definition already takes special cases into account.

Term 5: Element

Elements are:

0. Omni (also Empty for aura)
1. Pyro

2. Hydro
3. Electro
4. Cryo
5. Dendro
6. Anemo
7. Geo

Term 6: Damage Type

Damage Types not only include **Element** (Term 5), but also two additional options: piercing and physical. Omni damage has no practical meaning here.

Term 7: Energy

Energy is a non-negative integer.

Term 8: Dice

Dice is a pair: (**Element** (Term 5) or Black or White, Number)

We use term **Elemental Dice** to refer dice with concrete **Element** (Term 5).

Term 9: Dice Pattern

Dice Pattern is an unordered list of **Dice** (Term 8).

An example is [*Pyro*, 1, *Black*, 2], which is a typical setting for Normal Attack of Pyro character.

Term 10: Skill

A skill consists of

1. Skill Name. An identifier(string).
2. **Damage Type** (Term 6).
3. **Energy** (Term 7).
4. **Dice Pattern** (Term 9).

Term 11: Usage

Usage is a non-negative integer or positive infinity. It usually indicates how many times some effect can be triggered. For a permanent effect, it's positive infinity.

¹https://en.wikipedia.org/wiki/Genius_Invokation_TCG

Term 12: Superscript

Superscript is a **Usage** (Term 11) annotated in the upper right corner of some entity in game.

Term 13: Subscript

Subscript is a **Usage** (Term 11) annotated in the lower right corner of some entity in game.

Term 14: Zone

A zone contains a name, a set of units, and corresponding constraint rules. Units within a zone are necessarily identified by a subscript. Each unit's subscript is unique within its corresponding zone.

Term 15: Character Zone

Character Zone consists of:

1. Player ID
2. K characters
3. Active character index
1. Each character can possess a set of equipment.
2. Each character has an aura. An aura is a list of **Element** (Term 5)s. Omni represents none.
3. Each character has its own character status.
4. Each character has its own skill usage history.
 - (a) This history describes how many times each skill (distinguished by skill name) has been used in the current round.
 - (b) It also records how many times each skill has been used since the start of the game.
5. The active character possesses an active status. The active status belongs to the Character Zone (not to the character).

Term 16: Support Zone

Support Zone consists of:

1. Player ID
2. Up to 4 ordered support entities

Each support entity contains a name and a **Superscript** (Term 12).

Term 17: Summon Zone

Summon Zone consists of:

1. Player ID
2. Up to 4 ordered summons

Each summon contains a name, a **Damage Type** (Term 6), and a **Superscript** (Term 12).

Term 18: Dice Zone

Dice Zone contains K Elemental Dice (Term ??). For the opponent, only K is visible information in the Dice Zone.

Term 19: Hand Zone

Hand Zone consists of:

1. K cards in hand (maximum 10)
2. For the opponent, only K is visible information in the Hand Zone

Each card in hand includes a name, energy cost, and Elemental Dice (Term ??).

Term 20: Deck

Deck consists of:

1. M face-down cards. Only God knows the internal order.
2. For the current player, the deck contains a list of pairs in the form (card name, position)
 - (a) Position -1 indicates its position is unknown, or 0 M-1. 0 represents the bottom of the deck, M-1 represents the top of the deck.

Term 21: Card Graveyard

Graveyard consists of:

1. Player ID
2. An ordered list of tuples (sorted by timing)

The tuple format is (card name, timing). Timing is a string in the format of "Round-Turn", indicating when the card was played.

Term 22: Global Identifier

Each unit is determined by a route.

Term 23: Routing Rules

Use " " for separation. The root path is the player ID, followed by the zone ID, then the target subscript (subscript within the zone). For most units, three-level addressing is sufficient. However, for active status and character-affiliated attributes, this is still not enough.

1. Character Zone
 - 1.1 Character Zone

- 1.1.x A certain character
 - 1.1.x.1 Aura
 - 1.1.x.2 Equipment
- 1.2 Active Status

Glossary

1	Player	1
2	Current Player	1
3	Round	1
4	Turn	1
5	Element	1
6	Damage Type	1
7	Energy	1
8	Dice	1
9	Dice Pattern	1
10	Skill	1
11	Usage	1
12	Superscript	2
13	Subscript	2
14	Zone	2
15	Character Zone	2
16	Support Zone	2
17	Summon Zone	2
18	Dice Zone	2
19	Hand Zone	2
20	Deck	2
21	Card Graveyard	2
22	Global Identifier	2
23	Routing Rules	2